

Datasheet for BBO Capstone project

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Introduction

There are eight synthetic functions in the Black Box Optimisation (BBO) Capstone project. Each function is designed to mimic real-world complexity, with features such as non-linearity and multiple local maxima.

Motivation

A set of initial data has been provided for each BBO function. The initial data set provides inputs to the real function and the corresponding outputs. The initial data set is provided as an input to build a surrogate model.

Composition

The initial data set contains random inputs to the real function and its output. Every week a new candidate input point is run through the real function and the output is added to the initial data set and then the cycle (surrogate → sample points → acquisition function) is repeated with the full (initial + new data) dataset.

Collection process

A surrogate model is developed using the dataset, then a set of sample points is chosen using different techniques like Cartesian grid, Latin hypercube sampling (lhs) and Sobol, the surrogate model is used to predict the outputs for the chosen sample points. Then an acquisition function is used to select the next point to sample by balancing exploration of uncertain regions and exploitation of areas predicted to have promising objective values, focusing effort where it is most beneficial and making optimisation more efficient. The best candidate point is provided as an input to the real function. The new point and the output of the real function is then added to the initial data set. The data set grows by one input and output each week for the duration of the project.

Preprocessing and uses

The input and output data for all functions is numerical and is transformed so that it shares a common scale before it is used for any processing.

The preprocessing steps identify outlier inputs/outputs and removes them from the dataset.

The data set does not contain any data that can be categorised as demographic, ethnic or personally identifiable information (PII) therefore cannot be used inappropriately.

Distribution and maintenance

The data set is available from the GitHub repository location at <https://github.com/ajay-falcon-ai/Capstone-Black-Box-functions>

The data set will be maintained by the admin of the GitHub repository and will be updated throughout the duration of the Capstone project.